Customer Shopping Behavior Analysis

Deep-dive analysis of 3,900 retail transactions to identify core drivers of revenue and customer loyalty.

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Project Overview

Objective

Transform raw CSV data into structured insights using Python, SQL (PostgreSQL), and Power BI to answer critical business questions about demographics and spending behavior.

Scope

Analysis of 3,900 retail transactions with focus on revenue drivers, customer segments, and loyalty patterns.

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Data Engineering & Schema

Cleaned data in Python for SQL compatibility. Final PostgreSQL schema includes key attributes for analysis.

customer_id

Unique identifier for each shopper

age

Customer age for demographic grouping

purchase_amount_usd

Total transaction value

review_rating

Customer feedback score (1.0-5.0)

subscription_status

Member status (Yes/No)

gender

Male/Female classification

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Revenue Contribution by Gender

SQL Query

```
SELECT gender, SUM(purchase_amount_usd)...
```

Key Finding

Male customers contribute significantly higher total revenue compared to **Female** customers in this dataset.

Business Impact

Optimize marketing budgets to target highest-spending gender while creating "Discovery" campaigns for lower-spending segment.

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Average Spend by Age Group

1

Query Structure

SQL CASE statement groups ages into 18-24, 25-40, 41-60, and 60+ segments

2

Key Finding

25-40 and **41-60** groups show highest average transaction value

3

Business Impact

Target "Prime Spenders" with premium product bundles to increase average order value

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Customer Loyalty Segmentation

SQL Logic

```
CASE WHEN
previous_purchases = 1
  THEN 'New'
  ELSE 'Loyal'
END
```

Key Finding

Majority are **Loyal Customers**, but **New Customer** segment is growing.

Business Impact

Focus on "Referral Program" for Loyal segment to lower cost of acquiring new customers.

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Power BI Dashboard

PostgreSQL database connected to Power BI for live, interactive visualization.

Key KPI Cards

Total Revenue, Total Customers (3,900), Average Review Rating

Donut Charts

Payment Method distribution across transactions

Bar Charts

Revenue by Category for performance tracking

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Final Recommendations

01

Retention Focus

Implement "VIP Early Access" program for loyal segments to maintain engagement

02

Payment Incentives

Promote Credit Card usage for high-ticket items to reduce checkout friction

03

Category Expansion

Direct marketing toward top-performing categories to maximize ROI on ad spend

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Key Takeaways

Male Customers Drive Revenue

Higher total revenue contribution from male shoppers

Prime Spenders: 25-60

25-40 and 41-60 age groups show highest average transaction value

Loyal Base Growing

Majority are loyal customers with growing new customer segment

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